

Aufgabe 24

Binomialverteilung

A bar chart showing the probability mass function for a binomial distribution with $n=3$. The x-axis is labeled k and ranges from 0 to 8. The y-axis is labeled $P(X_3 = k)$. The bars are located at $k=0, 1, 2, 3, 4, 5, 6, 7, 8$ with heights corresponding to the binomial coefficients $\binom{3}{k}$. <table border="1"><thead><tr><th>k</th><th>$P(X_3 = k)$</th></tr></thead><tbody><tr><td>0</td><td>$\frac{1}{8}$</td></tr><tr><td>1</td><td>$\frac{3}{8}$</td></tr><tr><td>2</td><td>$\frac{3}{8}$</td></tr><tr><td>3</td><td>$\frac{1}{8}$</td></tr><tr><td>4</td><td>0</td></tr><tr><td>5</td><td>0</td></tr><tr><td>6</td><td>0</td></tr><tr><td>7</td><td>0</td></tr><tr><td>8</td><td>0</td></tr></tbody></table>	k	$P(X_3 = k)$	0	$\frac{1}{8}$	1	$\frac{3}{8}$	2	$\frac{3}{8}$	3	$\frac{1}{8}$	4	0	5	0	6	0	7	0	8	0	☒
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A bar chart showing the probability mass function for a binomial distribution with $n=5$. The x-axis is labeled k and ranges from 0 to 8. The y-axis is labeled $P(X_5 = k)$. The bars are located at $k=0, 1, 2, 3, 4, 5, 6, 7, 8$ with heights corresponding to the binomial coefficients $\binom{5}{k}$. <table border="1"><thead><tr><th>k</th><th>$P(X_5 = k)$</th></tr></thead><tbody><tr><td>0</td><td>$\frac{1}{32}$</td></tr><tr><td>1</td><td>$\frac{5}{32}$</td></tr><tr><td>2</td><td>$\frac{10}{32}$</td></tr><tr><td>3</td><td>$\frac{10}{32}$</td></tr><tr><td>4</td><td>$\frac{5}{32}$</td></tr><tr><td>5</td><td>$\frac{1}{32}$</td></tr><tr><td>6</td><td>0</td></tr><tr><td>7</td><td>0</td></tr><tr><td>8</td><td>0</td></tr></tbody></table>	k	$P(X_5 = k)$	0	$\frac{1}{32}$	1	$\frac{5}{32}$	2	$\frac{10}{32}$	3	$\frac{10}{32}$	4	$\frac{5}{32}$	5	$\frac{1}{32}$	6	0	7	0	8	0	☒
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Ein Punkt für das richtige Ankreuzen.